

Fast Online Video Synopsis Based on Potential Collision Graph

Yi He, Zhiguo Qu, Changxin Gao, Nong Sang

Abstract—Video synopsis is a smart solution to fast browsing and retrieval of raw surveillance data, in which tube rearrangement plays a key role. However, conventional methods for tube rearrangement are based on minimizing a global energy function, which is computational intensive and time consuming. In this paper, we propose a novel tube rearrangement strategy for online video synopsis by analyzing collision relationship between tubes. A Potential Collision Graph (PCG) is constructed to represent the tubes and their potential collision relationship. Based on the PCG, tube rearrangement is achieved by filling the tubes into synopsis video in a deterministic way, which decreases computational complexity. Finally, we incorporate the proposed tube rearrangement into an online framework to generate video synopsis and validate its efficiency with extensive experiments.

Index Terms—potential collision graph, online tube rearrangement, video synopsis

I. INTRODUCTION

VIDEO synopsis aims to condense the long video into shorter synopsis for fast retrieval and review of raw surveillance data. It has been an active research topic in the surveillance video processing communities recently. Different from earlier approaches such as video fast forward [1]–[3], video abstraction [4]–[7], and video summarization [8]–[10], video synopsis takes the tube as the basic processing unit. It shortens the video by appropriately rearranging the tubes in timeline, which preserves the activities and dynamics of moving objects completely.

The method proposed by Pritch *et al.* [11] is a pioneering work in this field, which includes an online phase and a response phase. The online phase includes tube extraction and background images recording whereas the response phase rearranges the tubes in timeline to determine display time of the tubes in the synopsis. Under this framework, several issues with respect to collision, speed and consistency have been considered. In [12], the synopsis video generation is formulated as a maximum a posteriori (MAP) estimation problem, where

the appearing frames of tubes are chronologically rearranged in real-time according to an online updated synopsis table. Another excellent online framework was proposed by Zhu *et al.* [13] which transforms the tube rearrangement into a stepwise optimization problem and uses multicore technique to further improve the speed. It is proven to be suitable for real-time endless surveillance videos. Similar online strategy is reported in [14], which takes into consideration motion structure between tubes. Other methods try to improve video synopsis from different aspects, such as shifting tubes in temporal-spatial domain [15], adopting novel tracking strategies [16], extracting perfect tubes [17], [18], performing synopsis on abnormal tubes to create more concise synopsis [19], seam carving [20], [21], and scaling down objects [22].

The procedures of video synopsis include tube extraction, background generation, tube rearrangement, and tube stitching, for which the tube rearrangement plays a key role in the whole process. In the aforementioned methods, tube rearrangement is achieved by iteratively minimizing a global energy function. It consists of two terms: 1) an unary energy term accounts for factors such as maximum activity and consistency with background and 2) a pairwise energy term accounts for chronological order and collisions artifacts. Obviously, minimizing this global energy function is time-consuming when the number of tubes is huge. Besides, it is memory demanding to store all tubes and backgrounds for a long intact video. In order to address this issue, the tube rearrangement is usually implemented in an offline manner. To enable online implementation of tube rearrangement, [13], [14] proposed to break the long intact videos down into pieces and achieve tube rearrangement through stepwise optimization, which improves the speed significantly. Moreover, they leave out the unary costs because these factors are not significant in the online framework [13]. However, iterative computation of collision costs is still needed during optimization, which causes large amounts of computation redundancy.

In this paper, we propose a novel online tube rearrangement strategy to avoid the iterative computation of collision costs in the work [13], [14]. The key observation is that the potential collision relationship between tubes (*e.g.* whether or not the tubes will collide in the synopsis) can be determined in advance. Accordingly, we define two kinds of relationship, *i.e.* *Collision Free (CF)* and *Collision Potential (CP)* to describe the relationship between tubes and construct a Potential Collision Graph (PCG) to represent it. Based on the PCG, we determine the starting-time labels of tubes in the synopsis in a deterministic way instead of using any time-consuming optimization. Such strategy computes the collision costs only

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once and thus accelerates the tube rearrangement procedure. Finally, we incorporate it into the online framework [13], [14] to demonstrate its efficiency.

The rest of the paper is organized as follows. Section II presents the details of tube rearrangement. Section III introduces the flowchart of online synopsis. Section IV shows the experimental results and Section V gives the conclusions.

II. TUBE REARRANGMENT

Since our tube rearrangement strategy is proposed for online synopsis framework, we only consider the collision costs between tubes and leave out other unary terms (e.g., activity cost and background inconsistency) as researchers did in [13].

A. Relationship between Tubes

The underlying reason for unpleasant collisions phenomenon in the synopsis is that two tubes are rearranged to pass the same area at the same time. On the contrary, if the two tubes pass the colliding area at different times in the synopsis, their collision can be avoided. Based on this observation, we define two types of relationship to determine the colliding possibility between tubes before rearranging them.

1) *Collision Free (CF)*: Two tubes are said to be *Collision free (CF)* if they never occupy the same area after being projected onto the spatial domain as shown in Fig.1(a) and Fig.1(b). One *CF* example is shown by the relationship of tube A with any other tube in Fig.1(b). In this case, the two tubes will never cause collision in synopsis at all when being shifted along the time axis. As a result, tubes with this relationship can be rearranged free of collisions.

2) *Collision Potential (CP)*: Two tubes are said to be *Collision potential (CP)* if their part of trajectories ever share the same area after being projected onto the spatial domain. The *CP* includes two different cases: colliding in the same direction (*CSD*) and colliding in the opposite direction (*COD*). The same/opposite direction means that the angle between the tubes' moving directions is in $[0, \pi]$ and $[\pi, 2\pi]$ respectively. These two cases are the basic reason of causing serious collisions in the synopsis. For the *CP* tubes, we record the corresponding starting and ending frames of the collision parts between them for subsequent use. For example, let t_k^{cs} and t_k^{ce} represent the starting and ending frame of colliding part in tube k , where $k = B, C, D, E$ in Fig.1(b). In the case of *CSD* as shown by tube D and tube E in Fig.1(b), t_D^{cs} and t_D^{ce} correspond to t_E^{cs} and t_E^{ce} respectively, which are recorded as $t_D^{cs} \leftrightarrow t_E^{cs}$ and $t_D^{ce} \leftrightarrow t_E^{ce}$. On the contrary, in the case of *COD* as shown by tube B and tube C in Fig.1(b), the corresponding relations are recorded as $t_B^{cs} \leftrightarrow t_C^{ce}$ and $t_B^{ce} \leftrightarrow t_C^{cs}$.

The *CF* and the *CP* relationships between tubes can be simply determined by computing the overlapping areas of their bounding boxes in the spatial domain. In the *CP* relationship, the cases of *CSD* and *COD* can be differentiated by checking the chronological order of collision part between tubes, which can be seen in Fig.1(b).

B. Potential Collision Graph

We use an undirected attribute graph called Potential Collision Graph (PCG) to represent the tubes and their potential collision relationship, as shown in Fig.1(c) and 1(d). In the

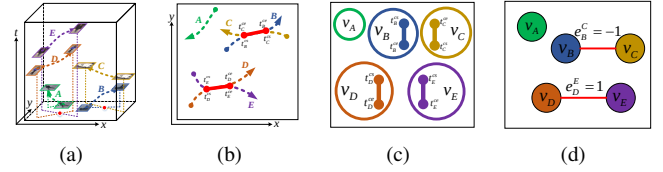


Fig. 1. Five spatial-temporal tubes in (a) are mapped onto the spatial domain in (b) for analyzing potential collision relationships between tubes. We define two type of relationship between tubes: *collision free* and *collision potential*. The potential collision relationships of five tubes in (b) are abstracted in (c) where each vertex stands for a tube. The edges are connected if collision potential relationships exist to construct the PCG in (d).

PCG, the vertices stand for tubes extracted from the video and the edges indicate potential collision relationship between tubes. The attributes of vertices record the information about the tube and the potential collision time of the tube with other tubes. The vertices standing for *CP* tubes are connected by edges with weights 1 and -1 in Fig.1(d), indicating the *CSD* and *COD* cases respectively. That is, $e_D^E = 1$ and $e_B^C = -1$ in Fig.1(d). The vertices standing for *CF* tubes are left unconnected and become isolated vertices in the PCG, as shown by Fig.1(d). For convenience, we use an adjacency matrix M to store the PCG.

C. Tube Rearrangement

Tube rearrangement aims to fill the tubes into the synopsis video by labeling their starting frames. Different from earlier work [11]–[15], we implement the tube rearrangement by filling the tubes into a temporal condense space L_{temp} one by one. The $L_{temp} = \{1, 2, \dots\}$ is a label set that temporarily stores the incoming tubes before filling them into the final synopsis.

Once a N tubes subset $Q = \{T_i\}_{i=1}^N$ is obtained, we construct a PCG $G = \{v_i, e_{ij}\}_{i,j=1}^N$ and an adjacency matrix M of size $N \times N$ to represent the relationships (see Fig.2). The vertices in G are numbered according to the first-appearing frame of the tubes, which are used as the processing order here. The task now is to determine the starting-time labels l_i , $i = 1, 2, \dots, N$ in L_{temp} according to the PCG.

Initially, we shift the first tube to the beginning of L_{temp} , i.e., $l_1 = 1$. Then we label other tubes in Q sequentially one by one. Let T_i denotes the tube in Q to be rearranged and T_j stands for the already-labeled tube in L_{temp} , where $2 \leq i \leq N$ and $1 \leq j \leq i$.

1) **CP Case.** If $M_{ij} = 1$, $j \in [1, \dots, i]$, it means T_i and T_j are *CSD*. The corresponding starting and ending frames of collision part between tube T_i and T_j are recorded as $t_i^{cs} \leftrightarrow t_j^{cs}$ and $t_i^{ce} \leftrightarrow t_j^{ce}$. To prevent the potential collision from becoming real ones, one should ensure that the incoming tube T_i go through the potential collision part with tube T_j at different time. As shown by the *CSD* case in Fig.2, the tube T_i should be rearranged in R_{i1} or R_{i3} rather than R_{i2} to avoid collision. Assume $l_i^{cs} \leftrightarrow l_j^{cs}$ and $l_i^{ce} \leftrightarrow l_j^{ce}$ denote the starting and ending label of collision part between T_i and T_j , the constraint can be expressed as:

$$(l_i^{cs} - l_j^{cs})(l_i^{ce} - l_j^{ce}) \geq 0 \quad (1)$$

Considering the sizes and the speeds of the colliding objects, one frame separation between potential collision parts may still cause minor collisions. We thus add a safety margin q (the pink area beside the collision zone in Fig.2) for ensuring

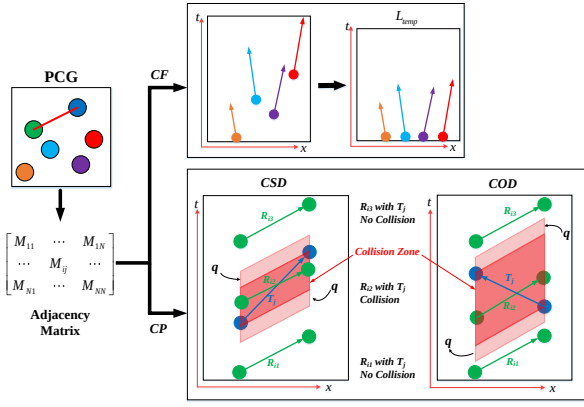


Fig. 2. Illustration of the tube rearrangement according to the PCG. The *CF* tubes are shifted to the first frame in L_{temp} . For the *CP* tubes, the incoming tube should be rearranged outside of the temporal collision zone of the already-labeled ones.

sufficient separation between their corresponding starting and ending frames of potential collision parts:

$$\min(|l_i^{cs} - l_j^{cs}|, |l_i^{ce} - l_j^{ce}|) \geq q \quad (2)$$

The setting of parameter q should take into account the sizes and the speeds of tubes in the scenario and will be discussed in the section IV.

The constraints in Eq.(1)-(2) can be easily imposed on the starting-time label l_i of tube T_i because $l_i^{cs} = l_i + (t_i^{cs} - t_i^s)$, $l_i^{ce} = l_i + (t_i^{ce} - t_i^s)$, $l_j^{cs} = l_j + (t_j^{cs} - t_j^s)$, and $l_j^{ce} = l_j + (t_j^{ce} - t_j^s)$, where t_i^s and t_j^s are the appearing time of tube T_i and T_j in the original video. All the allowable labels satisfying Eq.(1)-(2) for tube T_i are denoted as $L_i = \{l_i\}$.

If $M_{ij} = -1$, $j \in [1, \dots, i]$, it means T_i and T_j are *COD*. The constraints of *COD* are almost the same as *CSD* except that the starting and ending frames of collision parts between the T_i and T_j correspond reversely, i.e. $t_i^{cs} \leftrightarrow t_j^{ce}$ and $t_i^{ce} \leftrightarrow t_j^{cs}$, so the constraints change to:

$$(l_i^{cs} - l_j^{ce})(l_i^{ce} - l_j^{cs}) \geq 0 \quad (3)$$

$$\min(|l_i^{cs} - l_j^{ce}|, |l_i^{ce} - l_j^{cs}|) \geq q \quad (4)$$

If there are more than one $M(i, j) \neq 0$, $j \in [1, \dots, i]$, it means that T_i is *CP* with more than one tube. For each $M(i, j) \neq 0$, $j \in [1, \dots, i]$, we can obtain a label-set L_i^j , including all the possible temporal locations of T_i which will not collide with T_j . To condense the synopsis to a large extent, the start-time label for tube T_i should be as small as possible. We thus choose the minimum value from the intersection set of all the label-sets:

$$l_i = \min\left\{\bigcap_{j=1}^i L_i^j\right\} \quad (5)$$

2) **CF Case.** If $M_{ij} = 0$, $j \in [1, \dots, i]$, it means the tube T_i is *CF* with the already-labeled tubes. We put T_i in the first frame without worrying about collision happening. In other words, tube T_i is labeled by $l_i = 1$, as shown in the *CF* case in Fig.2.

D. Computational Cost Analysis

Obviously, the proposed tube rearrangement determines the tubes' starting-time labels in a deterministic way instead of using any time-consuming optimization process. It computes the collision relationship between tubes only once and eliminates

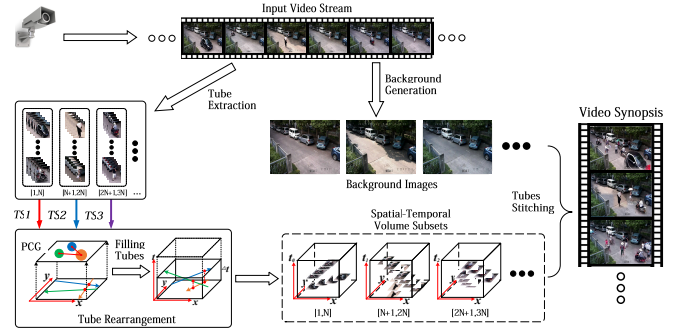


Fig. 3. Overview of our online framework. Once N tubes are extracted from input video stream, these tubes will be rearranged as a spatial-temporal volume subset. The synopsis video is generated by running the procedure and stitching all the volume subsets into background images.

the iterative computation of collision costs in conventional methods. Assume that the computational cost of collision costs in traditional methods is c , which roughly equals the computational cost of relationship determination (II.A) in our method. The number of tubes to be filled is N and the starting-time label set of tube filling is $\{1, 2, \dots, n\}$. The computational cost of our tube rearrangement strategy can be approximated as $N(N-1)c$, while the computational costs of tube rearrangement in [11] and [14] are $(n/k)^N N(N-1)c$ and $N(N-1)nc/2$, where k stands for jump steps between labels. Therefore, the computational ratio of the proposed tube arrangement to the methods in [11] and [14] are $(n/c)^N$ and $n/2$ respectively.

III. ONLINE SYNOPSIS GNERATION

Similar to [13], [14], we incorporate the proposed tube rearrangement into the online framework to generate video synopsis. It consists of four stages: tube extraction, background generation, tube rearrangement, and tube stitching, as shown in Fig.3. The whole process of online videos synopsis works as follows: Firstly, the background images are generated and objects are tracked to extract N tubes from the original video. Secondly, tube rearrangement is implemented to determine the starting-time labels of these N tubes to form the spatial-temporal volume subset. Finally, the spatial-temporal volume subset are stitched into background according to the starting-time labels determined in the former stage. Details in the online framework except the tube rearrangement stage are described as follows.

For background generation and tube extraction, we use the Visual Background Extractor (ViBe) algorithm [23]. ViBe is an efficient and universal sample-based background subtraction algorithm, which compares each pixel value with the neighbor values in a sphere to determine whether that pixel belongs to the background. The extracted objects are then tracked using the linear neighbor object prediction algorithm [23], [24] extract tubes. A tube that is not updated for more than 5 frames is considered terminated and will be buffered into tube subset for rearrangement. To solve the background images updating and long tube division problem, we adopt the method in [13].

For the tube stitching stage, we used the modified image editing method in [11], [25] to fuse the rearranged tubes into background according to the corresponding start-time labels.

TABLE I
FOUR SURVEILLANCE VIDEO IN OUR EXPERIMENT.

Input Video	Resolution	T_I	N_O	Relationships among Objects	T_S
Street	320*240	30264	547	CF(3876) CP(6220)	783
Crossroad	320*240	4384	478	CF(6226) CP(2743)	523
Gate	320*240	45994	330	CF(3700) CP(2446)	686
Road	320*240	38329	132	CF(910) CP(1499)	423

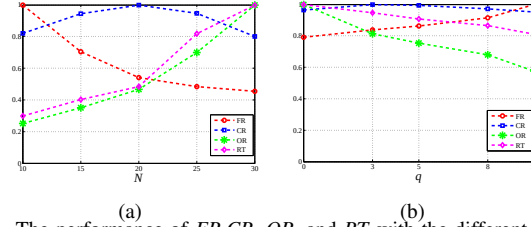


Fig. 4. The performance of *FR*, *CR*, *OR*, and *RT* with the different N and q . The results are average performance on 20 different videos and are normalized by respective maximum value.

Discontinuity of motion flow problem, which is caused by stepwise filling of tube subsets, is worth to mention here. We used the method [14] to solve this problem.

IV. EXPERIMENTAL RESULTS

In order to evaluate the performance of our proposed method, four surveillance video sequences taken from indoor and outdoor scenes are used, including scenarios of street, crossroad, gate, and road. In these scenarios, the moving objects contain pedestrians, motorbikes, cars, and so on, among which different relationships such as *CF* and *CP* exist. The characteristics of the video sequences are listed in Table I, where T_I and T_S are the number of input and synopsis frames, and N_O is the number of tubes in the video. The experiments are running on an Inter Core i5 computer with 2.66 GHz CPU and 4-GB memory.

For comparison, we implement three other online methods in [12]–[14] and an offline method in [11]. For [13], no hardware acceleration or multithread techniques is adopted for fair comparison. Four metrics [12] are used to evaluate the performance of three methods: Frame Condensation Ratio (*FR*), Compact Ratio (*CR*), Overlap Ratio (*OR*), and **Running Time (*RT*)** for tube rearrangement in milliseconds per frame.

For setting of the parameters N and q , we conduct experiments on 20 videos. The average performance with different N and q are shown in Fig.4. From Fig.4(a), we can see that *FR*, *OR* and *RT* varies monotonically with N , meaning that larger N gives better performance on *FR* but worse *OR* and *RT*. The *CR* gets the maximum value when $N = 20$. The results in Fig.4(b) are well consistent with our intuition. Larger q gives better performance in *OR* but worse in *CR* and *FR*. Compromising between the performance with different N and q , we set $N = 20$ and $q = 3$ in the comparison experiments with other methods.

Fig.5 shows the performance comparison between the proposed method with [11]–[14] and offline PCG (all the tubes are processed together instead of using the stepwise manner). It can be seen that our proposed method achieves comparable performance in *FR* with other online methods but worse than offline PCG. The reason is that the online method suffers from discontinuity between tube subsets, which is a common problem in online frameworks [13], [14]. Moreover, the proposed

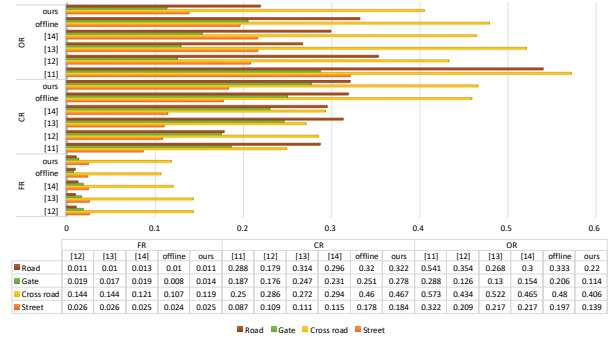


Fig. 5. Performance comparison using *FR*, *CR*, and *OR*.

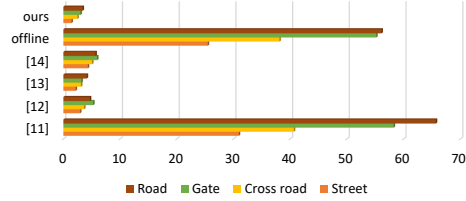


Fig. 6. *RT*(ms) comparison between the online and offline synopsis.



Fig. 7. Example frames in the video synopsis for different scenes. The upper 4 frames are original frames, and the lower 4 frames are the corresponding synopsis frames.

method achieves best performance on *CR* and *OR* among the five methods. It means that our method can better deal with collision phenomenon when rearranging the tubes. This is because our tube rearrangement strategy takes the collision cost as a hard constraint, which makes sure that *CP* tubes pass the colliding area at different times in the synopsis.

As shown in Fig.6, the speed advantage of the proposed method is obvious, especially significant comparing to the offline method. This is due to the elimination of iterative computation of collision cost in our method.

As for subjective evaluation, example frames of the synopsis videos for four scenes are shown in Fig.7. One can see, more objects from different time are stitched into the same frame with the key object (*i.e.* silver car, scrubwoman, bicycle, man with umbrella) in the video synopsis.

V. CONCLUSION

In this paper, we have proposed a novel approach to rearrange tubes in video synopsis, which enables online implementation of video synopsis. Unlike conventional methods for tube rearrangement based on minimizing a global energy function, our method rearrange tubes by analyzing their potential collision relationships. We represent the tubes and their relationship in the PCG, based on which we fill the tubes one by one in a deterministic way that eliminates the redundant computation of iterative collision costs in the optimization process. Experimental results have shown that the proposed method can generate more compact synopsis video with less collision artifacts and runs faster.

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